

Quantum Communications and Networks(QU.50013)

Exercise Set 2

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Problem 1

(a)

$$\rho_{AB} = \begin{pmatrix} \cos^2 \theta & 0 & 0 & \sin \theta \cos \theta \\ 0 & 0 & 0 & 0 \\ \sin \theta \cos \theta & 0 & 0 & \sin^2 \theta \end{pmatrix}$$
$$\Rightarrow \rho_A = \text{Tr}_B \rho_{AB} = \begin{pmatrix} \text{Tr} \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & 0 \end{pmatrix} & \text{Tr} \begin{pmatrix} 0 & \sin \theta \cos \theta \\ 0 & 0 \end{pmatrix} \\ \text{Tr} \begin{pmatrix} 0 & 0 \\ \sin \theta \cos \theta & 0 \end{pmatrix} & \text{Tr} \begin{pmatrix} 0 & 0 \\ 0 & \sin^2 \theta \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & \sin^2 \theta \end{pmatrix}$$

(b) ρ_B is in its diagonal form in the computational basis, so its eigenvalues are $\cos^2 \theta$ and $\sin^2 \theta$.

$$\Rightarrow S(\rho_A) = -\cos^2 \theta \log_2 \cos^2 \theta - \sin^2 \theta \log_2 \sin^2 \theta$$

(c) $S(\rho_A)(\theta = 0) = 0$, $S(\rho_A)(\theta = \pi/4) = 1$. In fact, the former corresponds to the classical state $|00\rangle$, while the latter corresponds to the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

(d) *If the global state is known to be pure*, then the fact that ρ_A describes a mixed state could allude to global entanglement, although no local operation nor measurement can distinguish a mixed state from a subsystem of an entangled state with identical reduced density matrices.

Problem 2

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